The Dot Product

ANSWER KEY

Section 1: Algebraic Definition and Properties

1 Find the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \langle 4, -3 \rangle$ and $\mathbf{v} = \langle 2, 5 \rangle$.

a. -7 ✓

b. 23

c. $\langle 8, -15 \rangle$

d. 1

$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 = (4)(2) + (-3)(5) = 8 - 15 = -7.$

2 What is the value of $\mathbf{v} \cdot \mathbf{v}$ for any vector $\mathbf{v}$?

a. $\|\mathbf{v}\|^2$ ✓

b. $\|\mathbf{v}\|$

c. 0

d. 1

$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 = \|\mathbf{v}\|^2.$

3 Which property does NOT hold for the dot product?

a. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (Commutative)

b. $c(\mathbf{u} \cdot \mathbf{v}) = (c\mathbf{u}) \cdot \mathbf{v}$

c. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$

d. $\mathbf{u} \cdot (\mathbf{v} \cdot \mathbf{w}) = (\mathbf{u} \cdot \mathbf{v}) \cdot \mathbf{w}$ **(Associative)** ✓

The dot product of two vectors is a scalar, so $\mathbf{v} \cdot \mathbf{w}$ is a scalar. You cannot dot a vector with a scalar, so associativity of dot products is undefined.

Section 2: Geometric Definition and Angle Between Vectors

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4 State the formula for the cosine of the angle θ between two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$.

a. $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$ ✓

b. $\cos\theta = \frac{\|\mathbf{u}\|\|\mathbf{v}\|}{\mathbf{u}\cdot\mathbf{v}}$

c. $\cos\theta = \mathbf{u}\cdot\mathbf{v}$

d. $\cos\theta = \frac{\mathbf{u}+\mathbf{v}}{\|\mathbf{u}+\mathbf{v}\|}$

From $\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$, we have $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$.

5 Find the angle between $\mathbf{u} = \langle 1, 1\rangle$ and $\mathbf{v} = \langle 0, 2\rangle$.

a. 45° ✓

b. 90°

c. 30°

d. 60°

$\mathbf{u}\cdot\mathbf{v} = 0 + 2 = 2$. $\|\mathbf{u}\| = \sqrt{2}$, $\|\mathbf{v}\| = 2$. $\cos\theta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$.

6 If the dot product $\mathbf{u}\cdot\mathbf{v} < 0$, what type of angle is θ?

a. **Obtuse (**$90^\circ < \theta \leq 180^\circ$**)** ✓

b. Acute ($0^\circ \leq \theta < 90^\circ$)

c. Right (90°)

d. Straight (180°)

Since $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$, a negative dot product implies $\cos\theta < 0$, which corresponds to an obtuse angle.

Section 3: Orthogonality and Parallel Vectors

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7 Two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular) if and only if:

a. $\mathbf{u} \cdot \mathbf{v} = 0$ ✓

b. $\mathbf{u} \cdot \mathbf{v} = 1$

c. $\mathbf{u} \times \mathbf{v} = \mathbf{0}$

d. $\|\mathbf{u}\| = \|\mathbf{v}\|$

$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos(90^\circ) = 0$.

8 Find k such that $\mathbf{u} = \langle 3, k \rangle$ is orthogonal to $\mathbf{v} = \langle 4, 6 \rangle$.

a. -2 ✓

b. 2

c. $-1/2$

d. 12

$\mathbf{u} \cdot \mathbf{v} = 3(4) + 6k = 12 + 6k = 0 \Rightarrow 6k = -12 \Rightarrow k = -2$.

9 Which pair of vectors is parallel?

a. $\langle 2, -3 \rangle$ **and** $\langle -6, 9 \rangle$ ✓

b. $\langle 2, 3 \rangle$ and $\langle 3, 2 \rangle$

c. $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$

d. $\langle 4, 1 \rangle$ and $\langle -1, 4 \rangle$

$\langle -6, 9 \rangle = -3\langle 2, -3 \rangle$. A vector is parallel to another if it is a scalar multiple of it.

Section 4: Vector Projections and Work

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10 State the formula for the vector projection of $\mathbf{u}$ onto $\mathbf{v}$ ($\text{proj}_{\mathbf{v}}\mathbf{u}$).

a. $\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\right)\mathbf{v}$ ✓

b. $\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|^2}\right)\mathbf{u}$

c. $\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$

d. $(\mathbf{u}\cdot\mathbf{v})\mathbf{v}$

The vector projection is the scalar component $\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$ multiplied by the unit vector $\frac{\mathbf{v}}{\|\mathbf{v}\|}$, yielding $\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\right)\mathbf{v}$.

11 Find $\text{proj}_{\mathbf{v}}\mathbf{u}$ for $\mathbf{u} = \langle 6, 2\rangle$ and $\mathbf{v} = \langle 4, 0\rangle$.

a. $\langle 6, 0\rangle$ ✓

b. $\langle 4, 0\rangle$

c. $\langle 0, 2\rangle$

d. $\langle 24, 0\rangle$

$\mathbf{u}\cdot\mathbf{v} = 24$. $\|\mathbf{v}\|^2 = 16$. $\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{24}{16}\langle 4, 0\rangle = \frac{3}{2}\langle 4, 0\rangle = \langle 6, 0\rangle$.

12 A force $\mathbf{F} = \langle 30, 40\rangle$ N moves a cart along the displacement $\mathbf{d} = \langle 5, 0\rangle$ m. Calculate the work done.

a. 150 J ✓

b. 200 J

c. 350 J

d. 250 J

$W = \mathbf{F}\cdot\mathbf{d} = (30)(5) + (40)(0) = 150$ J.

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